

Error, Norm, and Convergence-Rate Computation in the Distributed Periodic Poisson-Bracket Maxwell Study

DiscoGMPI Technical Report

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1 Purpose and scope

This report documents, in mathematical and implementation terms, the diagnostics computed by the convergence driver

`examples/convergence_distributed_periodic_poisson_bracket_maxwell.jl`.

It covers:

- component-wise errors for $E_x, E_y, E_z, H_x, H_y, H_z$;
- aggregate electric, magnetic, and six-component field errors;
- relative field errors;
- quadrature-sampled L^∞ diagnostics;
- energy discrepancy and discrete energy drift;
- mesh-size measures;
- observed convergence rates;
- the final PASS/FAIL convergence verdict.

The report follows the current source code. It also identifies distinctions that are important when interpreting the CSV output:

1. The CSV distinguishes the requested and measured mesh sizes. `target_h` records the Gmsh request. Rates use the measured `characteristic_h`.
2. `energy_error` compares numerical and analytical energies at the final time, while `relative_energy_error` measures numerical energy drift from the initial numerical energy.
3. the reported L^∞ values are maxima over cubature points, not mathematically exact suprema over the continuous domain.
4. component rates for exact-zero components measure decay of numerical leakage and can become dominated by roundoff.

2 Relevant source-code path

The computation is split across the following files.

Source	Responsibility
<code>examples/convergence_distributed_periodic_poisson_bracket_maxwell.jl</code>	Mesh-family generation, time integration, component errors, sampled L^∞ errors, rate calculation, CSV writing, and PASS/FAIL logic.
<code>examples/distributed_periodic_poisson_bracket_maxwell.jl</code>	Analytical plane wave, nodal-to-cubature interpolation, aggregate field norms, analytical energy, charge, and related quadrature diagnostics.
<code>examples/meshes/periodic_box_unstructured.geo</code>	Periodic box geometry and the Gmsh parameter <code>NxTarget</code> , with $h_{\text{target}} = L_x/N_x$.
<code>src/solver/kernels/metrics.jl</code>	Affine reference-to-physical tetrahedron map and element Jacobian.

Source	Responsibility
<code>src/solver/kernels/reference_tet.jl</code>	Degree- N nodal tetrahedral basis, Vandermonde matrices, mass matrix, and reference derivative operators.
<code>src/solver/kernels/JaskowiecSukumar.jl</code>	Jaskowiec–Sukumar tetrahedral cubature points and weights.
<code>src/DistributedDG.jl</code>	Distributed mass-matrix Maxwell energy and MPI reduction over owned elements.

3 Analytical Maxwell problem

3.1 Material model and wave parameters

The convergence driver currently enforces

$$\varepsilon = 1, \quad \mu = 1,$$

although the analytical helper functions retain the general symbols ε and μ . The wave speed and dispersion relation are

$$c = \frac{1}{\sqrt{\varepsilon\mu}}, \quad \omega = kc = \frac{k}{\sqrt{\varepsilon\mu}}.$$

The magnetic amplitude is $H_0 = 1$.

Let $x_0 = x_{\min}$ be the lower x -coordinate inferred from the mesh. The phase is

$$\theta(x, t) = k(x - x_0) - \omega t.$$

The analytical fields used for initialization and final-time error measurement are

$$\mathbf{E}^{\text{ex}}(x, y, z, t) = \left(0, 0, -\frac{kH_0}{\omega\varepsilon} \sin \theta \right), \quad (1)$$

$$\mathbf{H}^{\text{ex}}(x, y, z, t) = (0, H_0 \sin \theta, 0). \quad (2)$$

For the default $\varepsilon = \mu = H_0 = 1$, $\omega = k$, so

$$\mathbf{E}^{\text{ex}} = (0, 0, -\sin \theta), \quad \mathbf{H}^{\text{ex}} = (0, \sin \theta, 0).$$

The exact nonzero components are therefore E_z and H_y . The exact components E_x, E_y, H_x, H_z are identically zero. This fact is important when interpreting their component-wise errors and rates.

3.2 Periodic compatibility

For a box of length $L_x = x_{\max} - x_{\min}$, the driver requires

$$\frac{kL_x}{2\pi} \in \mathbb{N}.$$

Thus an integer number of wavelengths fits across the periodic x direction. Opposite faces in all three coordinate directions are paired, using translations

$$(-L_x, 0, 0), \quad (0, -L_y, 0), \quad (0, 0, -L_z).$$

4 Mesh family and mesh-size definitions

4.1 Gmsh target refinement

The Gmsh geometry defines

$$h_{\text{target}} = \frac{L_x}{N_x},$$

where N_x is the value of `NxTarget`. With a base target $N_{x,0}$, the convergence driver constructs

$$N_{x,\ell} = 2^\ell N_{x,0}, \quad \ell = 0, 1, \dots, L-1.$$

Therefore,

$$h_{\text{target},\ell} = \frac{L_x}{N_{x,\ell}} = 2^{-\ell} h_{\text{target},0}.$$

For the default four levels, this is exactly

$$h_0, \quad \frac{h_0}{2}, \quad \frac{h_0}{4}, \quad \frac{h_0}{8}.$$

Gmsh creates a new unstructured tetrahedral mesh for every level. The meshes are not required to be nested. `target_h` represents the requested Gmsh size, not the diameter of every tetrahedron.

4.2 Characteristic mesh size

The driver also reports

$$h_{\text{char}} = \left(\frac{|\Omega|}{N_{\text{el}}} \right)^{1/3},$$

where N_{el} is the global tetrahedron count. Element volume is calculated from the affine map:

$$|K| = |\widehat{K}| |\det J_K|, \quad |\widehat{K}| = \frac{4}{3}.$$

Each MPI rank sums only its owned element volumes. An `MPI.Allreduce(..., +, ...)` produces the global volume and global element count.

Important distinction. The convergence-rate denominator uses the measured, volume-derived size stored in `characteristic_h`. The requested Gmsh size is retained separately as `target_h`. On an unstructured Gmsh family the characteristic size need not decrease by exactly a factor of two, even though the requested target size does.

5 DG representation and evaluation at cubature points

5.1 Nodal polynomial field

On each tetrahedron K , every scalar component is represented by a degree- N nodal polynomial:

$$u_h|_K(\mathbf{x}) = \sum_{j=1}^{N_p} u_{K,j} \ell_j(F_K^{-1}(\mathbf{x})),$$

with

$$N_p = \frac{(N+1)(N+2)(N+3)}{6}.$$

The six component arrays are

$$E_x, E_y, E_z, H_x, H_y, H_z \in \mathbb{R}^{N_p \times N_{\text{local}}}.$$

5.2 Reference tetrahedron and physical map

The reference tetrahedron has vertices

$$(-1, -1, -1), \quad (1, -1, -1), \quad (-1, 1, -1), \quad (-1, -1, 1).$$

For reference coordinates (r, s, t) , the affine shape functions are

$$\phi_1 = -\frac{r+s+t+1}{2}, \quad \phi_2 = \frac{r+1}{2}, \quad (3)$$

$$\phi_3 = \frac{s+1}{2}, \quad \phi_4 = \frac{t+1}{2}. \quad (4)$$

The physical point is

$$\mathbf{x} = F_K(r, s, t) = \sum_{a=1}^4 \phi_a(r, s, t) \mathbf{x}_a.$$

The Jacobian is constant on each affine tetrahedron:

$$J_K = [(\mathbf{x}_2 - \mathbf{x}_1)/2 \quad (\mathbf{x}_3 - \mathbf{x}_1)/2 \quad (\mathbf{x}_4 - \mathbf{x}_1)/2].$$

Consequently,

$$d\mathbf{x} = |\det J_K| d\boldsymbol{\xi}.$$

5.3 Interpolation matrix

The solution is stored at DG interpolation nodes, while errors are evaluated at volume cubature points. The code constructs

$$I_q = V_q V^{-1},$$

where V is the nodal Vandermonde matrix and V_q evaluates the same orthonormal modal basis at cubature points. If $\mathbf{u}_K = (u_{K,1}, \dots, u_{K,N_p})^T$, then

$$u_h(\mathbf{x}_{K,q}) = (I_q)_{q,:} \mathbf{u}_K.$$

This interpolation is performed independently for all six field components.

6 Volume cubature

For DG order N , the convergence driver selects the Jaskowiec–Sukumar cubature order

$$p_q = \max(2, 2N + 4).$$

Let $\{(\boldsymbol{\xi}_q, w_q)\}_{q=1}^{N_q}$ be the resulting rule on $\widehat{K}$. A physical-domain integral is approximated by

$$\int_K f(\mathbf{x}) d\mathbf{x} \approx \sum_{q=1}^{N_q} |\det J_K| w_q f(F_K(\boldsymbol{\xi}_q)).$$

The implementation calls

$$W_{K,q} = |\det J_K| w_q$$

the *physical weight*.

The cubature rule exactly integrates polynomials up to its supported degree. However, the analytical wave contains sine functions. Therefore, quantities such as $(u_h - u^{\text{ex}})^2$ are not polynomials, and the reported norm is a high-order quadrature approximation to the continuous norm rather than a symbolically exact integral. The choice $2N + 4$ deliberately exceeds the degree needed for the square of a degree- N polynomial and reduces quadrature contamination from the trigonometric analytical field.

7 Component-wise L^2 errors

For each scalar component $u \in \{E_x, E_y, E_z, H_x, H_y, H_z\}$, define

$$e_u(\mathbf{x}, t_f) = u_h(\mathbf{x}, t_f) - u^{\text{ex}}(\mathbf{x}, t_f).$$

The continuous component norm is

$$\|e_u\|_{L^2(\Omega)} = \left(\sum_{K \in \mathcal{T}_h} \int_K e_u^2 d\mathbf{x} \right)^{1/2}.$$

The implemented cubature approximation is

$$\|e_u\|_{L^2(\Omega),q} = \left[\sum_{K \in \mathcal{T}_h} \sum_{q=1}^{N_q} W_{K,q} ((I_q)_q \cdot \mathbf{u}_K - u^{\text{ex}}(\mathbf{x}_{K,q}, t_f))^2 \right]^{1/2}.$$

The corresponding CSV fields are:

$$\begin{array}{lll} \text{12_ex_error,} & \text{12_ey_error,} & \text{12_ez_error,} \\ \text{12_hx_error,} & \text{12_hy_error,} & \text{12_hz_error.} \end{array}$$

7.1 Exact-zero components

For this plane wave,

$$E_x^{\text{ex}} = E_y^{\text{ex}} = H_x^{\text{ex}} = H_z^{\text{ex}} = 0.$$

Thus, for example,

$$\|e_{E_x}\|_{L^2} = \|E_{x,h}\|_{L^2}.$$

These norms measure spurious cross-component leakage. On an unstructured tetrahedral mesh, interpolating a function of x into the full three-dimensional degree- N polynomial space does not generally produce a polynomial that depends on x alone. Its transverse derivatives can therefore generate nonzero inactive field components during evolution. Those components can converge one order more slowly than the active components and can eventually dominate an aggregate vector norm. Their observed rates can also become unstable once their magnitudes approach floating-point roundoff. The physically active component errors $\|e_{E_z}\|_{L^2}$ and $\|e_{H_y}\|_{L^2}$ are generally the most direct component-level measures of plane-wave accuracy.

8 Aggregate electric, magnetic, and total errors

At a point, the vector errors are

$$\mathbf{e}_E = \mathbf{E}_h - \mathbf{E}^{\text{ex}}, \quad \mathbf{e}_H = \mathbf{H}_h - \mathbf{H}^{\text{ex}}.$$

Their squared Euclidean magnitudes are

$$|\mathbf{e}_E|^2 = e_{E_x}^2 + e_{E_y}^2 + e_{E_z}^2, \tag{5}$$

$$|\mathbf{e}_H|^2 = e_{H_x}^2 + e_{H_y}^2 + e_{H_z}^2. \tag{6}$$

The aggregate errors are

$$\mathcal{E}_E = \|\mathbf{E}_h - \mathbf{E}^{\text{ex}}\|_{L^2(\Omega)}, \tag{7}$$

$$\mathcal{E}_H = \|\mathbf{H}_h - \mathbf{H}^{\text{ex}}\|_{L^2(\Omega)}, \tag{8}$$

$$\mathcal{E}_{EH} = (\mathcal{E}_E^2 + \mathcal{E}_H^2)^{1/2}. \tag{9}$$

Because all terms use the same quadrature,

$$\mathcal{E}_E^2 = \|e_{E_x}\|_{L^2,q}^2 + \|e_{E_y}\|_{L^2,q}^2 + \|e_{E_z}\|_{L^2,q}^2, \quad (10)$$

$$\mathcal{E}_H^2 = \|e_{H_x}\|_{L^2,q}^2 + \|e_{H_y}\|_{L^2,q}^2 + \|e_{H_z}\|_{L^2,q}^2. \quad (11)$$

The CSV names are:

$$\text{12_electric_error} = \mathcal{E}_E, \quad \text{12_magnetic_error} = \mathcal{E}_H, \quad \text{12_total_error} = \mathcal{E}_{EH}.$$

9 Relative total field error

The analytical combined field norm is

$$\mathcal{N}_{EH}^{\text{ex}} = \left(\|\mathbf{E}^{\text{ex}}\|_{L^2,q}^2 + \|\mathbf{H}^{\text{ex}}\|_{L^2,q}^2 \right)^{1/2}.$$

The CSV quantity `relative_total_error` is

$$\mathcal{E}_{EH,\text{rel}} = \frac{\mathcal{E}_{EH}}{\max(\mathcal{N}_{EH}^{\text{ex}}, \epsilon_{\text{mach}})}.$$

The machine-epsilon guard prevents division by zero. For the nontrivial plane wave the denominator is not close to zero.

10 Quadrature-sampled L^∞ errors

The code also computes

$$\mathcal{E}_{E,\infty}^{(q)} = \max_{K,q} |e_E(\mathbf{x}_{K,q}, t_f)|, \quad (12)$$

$$\mathcal{E}_{H,\infty}^{(q)} = \max_{K,q} |e_H(\mathbf{x}_{K,q}, t_f)|, \quad (13)$$

$$\mathcal{E}_{EH,\infty}^{(q)} = \max_{K,q} \sqrt{|e_E(\mathbf{x}_{K,q}, t_f)|^2 + |e_H(\mathbf{x}_{K,q}, t_f)|^2}. \quad (14)$$

Each rank computes local maxima over owned elements. An MPI maximum reduction produces the global result.

The CSV fields are

$$\text{linf_electric_error}, \quad \text{linf_magnetic_error}, \quad \text{linf_total_error}.$$

Interpretation. These are not exact continuous L^∞ norms because extrema between cubature points are not searched. A more precise name would be “cubature-point maximum error.” The values become more representative as the mesh and sampling rule are refined, but they should not be presented as rigorous suprema.

11 Energy diagnostics

11.1 Analytical and numerical energy at final time

The electromagnetic energy density is

$$w(\mathbf{x}, t) = \frac{1}{2}\epsilon |\mathbf{E}|^2 + \frac{1}{2}\mu |\mathbf{H}|^2.$$

The quadrature diagnostic computes

$$\mathcal{H}_h(t_f) \approx \sum_{K,q} W_{K,q} \left[\frac{1}{2} \varepsilon |\mathbf{E}_h(\mathbf{x}_{K,q}, t_f)|^2 + \frac{1}{2} \mu |\mathbf{H}_h(\mathbf{x}_{K,q}, t_f)|^2 \right], \quad (15)$$

$$\mathcal{H}^{\text{ex}}(t_f) \approx \sum_{K,q} W_{K,q} \left[\frac{1}{2} \varepsilon |\mathbf{E}^{\text{ex}}(\mathbf{x}_{K,q}, t_f)|^2 + \frac{1}{2} \mu |\mathbf{H}^{\text{ex}}(\mathbf{x}_{K,q}, t_f)|^2 \right]. \quad (16)$$

The CSV field `energy_error` is the signed difference

$$\text{energy_error} = \mathcal{H}_h(t_f) - \mathcal{H}^{\text{ex}}(t_f).$$

It is not an absolute value and is not normalized.

11.2 Discrete energy drift

The convergence driver separately evaluates the numerical energy with the DG mass matrix:

$$\mathcal{H}_{M,h} = \frac{1}{2} \sum_K |\det J_K| \left[\varepsilon \sum_{\alpha=x,y,z} \mathbf{E}_{\alpha,K}^T M \mathbf{E}_{\alpha,K} + \mu \sum_{\alpha=x,y,z} \mathbf{H}_{\alpha,K}^T M \mathbf{H}_{\alpha,K} \right].$$

The CSV field `relative_energy_error` is

$$\text{relative_energy_error} = \frac{\mathcal{H}_{M,h}(t_f) - \mathcal{H}_{M,h}(0)}{\max(\mathcal{H}_{M,h}(0), \epsilon_{\text{mach}})}.$$

It measures conservation of the numerical Hamiltonian. It is not the relative version of `energy_error`. The current CSV naming should be interpreted with this distinction in mind.

12 Charge diagnostics

The source-free exact solution has zero electric and magnetic charge. The code evaluates the DG polynomial divergences and integrates

$$Q_E = \int_{\Omega} \varepsilon \nabla \cdot \mathbf{E}_h d\mathbf{x}, \quad (17)$$

$$Q_H = \int_{\Omega} \mu \nabla \cdot \mathbf{H}_h d\mathbf{x}. \quad (18)$$

The quadrature approximations are written as `electric_charge` and `magnetic_charge`. These are global signed integrals, not L^2 norms of divergence. Positive and negative local divergence errors can cancel.

13 MPI ownership and reductions

Every volume loop is restricted to `distributed_mesh.partition.owned`. Ghost elements are excluded from integration. This prevents interface halo copies from being counted more than once.

The reduction pattern is:

- squared-error integrals, energies, volume, element counts, and charges: `MPI.Allreduce(local, +, comm);`
- sampled L^∞ values: `MPI.Allreduce(local, max, comm);`
- stable time-step estimate: `MPI.Allreduce(local_dt, min, comm);`
- measured elapsed time: `MPI.Allreduce(local_elapsed, max, comm).`

All ranks therefore obtain the same global diagnostics. The CSV is written only by rank zero.

14 Time integration and its effect on measured error

For polynomial order N , the convergence test uses an H-first explicit symplectic partitioned Runge–Kutta scheme of order

$$p_t = N + 1.$$

Each rank estimates a stable local step. The global estimate is the minimum over ranks. The final number of steps and actual step are

$$n_{\text{step}} = \max \left(1, \left\lceil \frac{T}{\Delta t_{\text{est}}} \right\rceil \right), \quad \Delta t = \frac{T}{n_{\text{step}}}.$$

This guarantees that the numerical solution lands exactly at the requested final time T .

The measured field error is the combined result of:

1. spatial DG discretization error;
2. time-integration error;
3. volume-cubature error in the reported norm;
4. floating-point and MPI reduction error.

The choice $p_t = N + 1$ and a mesh-dependent CFL step is intended to prevent a low-order temporal method from dominating the spatial study. Nevertheless, a strict spatial-convergence investigation should repeat selected cases with a smaller CFL and verify that rates and errors are insensitive to further time-step reduction.

15 Observed convergence rates

15.1 Rate formula

For two consecutive mesh levels, let e_c be the coarse error and e_f the fine error. The code computes

$$p_{\text{obs}} = \frac{\log(e_c/e_f)}{\log(h_c/h_f)}.$$

Here

$$h_c = \text{characteristic_h}_c, \quad h_f = \text{characteristic_h}_f.$$

The measured ratio h_c/h_f is read from the two result rows. It is not assumed to equal two, because an independently generated unstructured Gmsh family generally follows the requested target sizes only approximately.

The first mesh level has no preceding level, so its rate is stored as missing and written as an empty CSV field. If either error is nonpositive, the helper also returns missing. In normal finite-precision calculations the norms are nonnegative and are positive unless the error is represented as exactly zero.

15.2 Rates stored in the CSV

The aggregate rates are:

$$\text{rate_electric}, \quad \text{rate_magnetic}, \quad \text{rate_total}.$$

The component rates are:

$$\begin{aligned} &\text{rate_ex}, \quad \text{rate_ey}, \quad \text{rate_ez}, \\ &\text{rate_hx}, \quad \text{rate_hy}, \quad \text{rate_hz}. \end{aligned}$$

Every rate is computed independently from the corresponding error column.

15.3 Meaning of rates for zero exact components

Rates for E_x, E_y, H_x, H_z describe how rapidly spurious components decay. While the leakage is resolved, these rates may be closer to N than to $N + 1$ and can lower the aggregate field rate. If the errors reach roundoff, their ratios can instead become erratic, extremely large, negative, or missing. The console prints E_z, H_y , and transverse-leakage diagnostics so that these regimes can be distinguished.

16 PASS/FAIL criterion

For each DG order N , the verdict uses only the final rate from the finest two mesh levels:

$$p_E^{\text{final}} = \text{rate_electric on the finest row,}$$

$$p_H^{\text{final}} = \text{rate_magnetic on the finest row.}$$

The strict target rates encoded in the driver are

$$p_E^{\text{expected}} = N + 1, \quad p_H^{\text{expected}} = N.$$

With the configured tolerance 0.5, an order passes if

$$p_E^{\text{final}} \geq N + 1 - 0.5, \tag{19}$$

$$p_H^{\text{final}} \geq N - 0.5. \tag{20}$$

Both values must be present and finite. The global verdict is PASS only if both tests pass for every requested DG order.

The $E = N + 1$ check is an enhanced aggregate-field target inherited from the aligned Poisson-bracket convergence experiment. It is not a generic theorem for every independently generated unstructured tetrahedral family. In particular, order- N transverse polarization leakage can make the full electric vector converge more slowly even when the active E_z component converges at order $N + 1$. A failed strict verdict must therefore be interpreted together with the component table rather than as evidence that the error integration is incorrect.

The following quantities do *not* currently affect PASS/FAIL:

- `rate_total`;
- all six component rates;
- monotonicity at every intermediate level;
- relative field error;
- L^∞ diagnostics;
- energy discrepancy or energy drift;
- charge diagnostics.

17 CSV data dictionary

Column	Definition
<code>mpi_ranks</code>	Number of MPI ranks in the run.
<code>boundary_condition</code>	Always <code>periodic</code> for this driver.

Column	Definition
order	DG polynomial degree N .
esprk_order	Time-integrator order $N + 1$.
cubature_order	Jaskowiec-Sukumar order $\max(2, 2N + 4)$.
mesh_level	Zero-based refinement level ℓ .
nx_target	Gmsh <code>NxTarget</code> value $N_{x,\ell}$.
target_h	$L_x/N_{x,\ell}$; requested Gmsh target size.
nelements	Global tetrahedron count.
min_owned_elements	Minimum owned-element count over ranks.
max_owned_elements	Maximum owned-element count over ranks.
characteristic_h	$(\Omega /N_{\text{el}})^{1/3}$; measured size used for rates.
dt	Actual step T/n_{step} .
nsteps	Number of time steps.
elapsed_seconds	Maximum time-integration wall time over ranks.
l2_electric_error	$\ \mathbf{E}_h - \mathbf{E}^{\text{ex}}\ _{L^2,q}$.
l2_magnetic_error	$\ \mathbf{H}_h - \mathbf{H}^{\text{ex}}\ _{L^2,q}$.
l2_total_error	$\sqrt{\mathcal{E}_E^2 + \mathcal{E}_H^2}$.
l2_ex_error	$\ E_{x,h} - E_x^{\text{ex}}\ _{L^2,q}$.
l2_ey_error	$\ E_{y,h} - E_y^{\text{ex}}\ _{L^2,q}$.
l2_ez_error	$\ E_{z,h} - E_z^{\text{ex}}\ _{L^2,q}$.
l2_hx_error	$\ H_{x,h} - H_x^{\text{ex}}\ _{L^2,q}$.
l2_hy_error	$\ H_{y,h} - H_y^{\text{ex}}\ _{L^2,q}$.
l2_hz_error	$\ H_{z,h} - H_z^{\text{ex}}\ _{L^2,q}$.
relative_total_error	$\mathcal{E}_{EH}/\mathcal{N}_{EH}^{\text{ex}}$, with an epsilon guard.
linf_electric_error	Maximum electric vector error at cubature points.
linf_magnetic_error	Maximum magnetic vector error at cubature points.
linf_total_error	Maximum combined six-component error at cubature points.
energy_error	Signed final quadrature energy difference $\mathcal{H}_h(t_f) - \mathcal{H}^{\text{ex}}(t_f)$.
relative_energy_error	Relative mass-matrix energy drift from $t = 0$ to t_f .
electric_charge	$\int_{\Omega} \varepsilon \nabla \cdot \mathbf{E}_h d\mathbf{x}$, by cubature.
magnetic_charge	$\int_{\Omega} \mu \nabla \cdot \mathbf{H}_h d\mathbf{x}$, by cubature.
rate_electric	Consecutive-level rate of <code>l2_electric_error</code> .
rate_magnetic	Consecutive-level rate of <code>l2_magnetic_error</code> .
rate_total	Consecutive-level rate of <code>l2_total_error</code> .
rate_ex, rate_ey, rate_ez	Consecutive-level rates of the three electric component errors.
rate_hx, rate_hy, rate_hz	Consecutive-level rates of the three magnetic component errors.

18 Consistency identities useful for auditing results

The following identities should hold up to floating-point summation error:

$$\text{l2.electric_error}^2 \approx \text{l2.ex_error}^2 + \text{l2.ey_error}^2 + \text{l2.ez_error}^2, \quad (21)$$

$$\text{l2.magnetic_error}^2 \approx \text{l2.hx_error}^2 + \text{l2.hy_error}^2 + \text{l2.hz_error}^2, \quad (22)$$

$$\text{l2.total_error}^2 \approx \text{l2.electric_error}^2 + \text{l2.magnetic_error}^2. \quad (23)$$

These are particularly useful for detecting component ordering mistakes or inconsistent analytical fields.

For the specified wave, one also expects:

- E_z and H_y dominate the field and error norms;
- E_x, E_y, H_x, H_z remain small;
- electric and magnetic aggregate errors are often similar because the exact wave has equal electric and magnetic energy for $\varepsilon = \mu = 1$;

- electric and magnetic charge integrals remain close to zero;
- relative energy drift remains small for the symplectic method.

19 Recommended execution and interpretation workflow

19.1 Run the convergence study

```
mpiexec -n 2 julia --project=. \
  examples/convergence_distributed_periodic_poisson_bracket_maxwell.jl \
  --nx-target=2 \
  --levels=4 \
  --orders=2,3,4 \
  --periods=1 \
  --cfl=0.05 \
  --output=output/convergence_distributed_periodic_poisson_bracket.csv
```

19.2 Interpretation order

For each DG order:

1. verify that `target_h` halves and that `characteristic_h` decreases consistently at every level;
2. verify that aggregate and active-component errors decrease;
3. check the three norm-consistency identities above;
4. inspect E_z and H_y rates;
5. inspect aggregate electric and magnetic rates;
6. treat zero-component rates as leakage diagnostics;
7. verify that a smaller CFL does not materially change the finest errors or rates;
8. inspect energy drift and charge as conservation checks;
9. use the automated PASS/FAIL verdict only after these checks.

20 Limitations and interpretation cautions

1. **Non-nested unstructured meshes.** The Gmsh family follows a uniform target-size sequence but is not a nested finite-element hierarchy. Pre-asymptotic rate variation is therefore expected.
2. **Target size versus actual sizes.** The requested value L_x/N_x is a mesher-control parameter. Rates use the measured volume-based `characteristic_h`, avoiding the false assumption that Gmsh exactly halves the realized element size.
3. **Space-time error coupling.** The final error includes temporal error. A CFL sensitivity check is necessary before attributing all observed behavior to spatial order.
4. **Quadrature approximation.** Trigonometric analytical fields make the norm integrands non-polynomial. The selected high-order cubature is accurate but not algebraically exact.
5. **Sampled L^∞ .** The maximum is taken only at volume cubature points.

6. **Charge cancellation.** Integrated signed charge can be small even when local divergence errors are not.
7. **Zero-component rates.** Exact-zero components can contain order- N polarization leakage before eventually becoming roundoff-dominated. They contribute to the aggregate vector norms.
8. **Energy column naming.** `energy_error` and `relative_energy_error` use different energy comparisons, as documented above.
9. **Verdict coverage.** PASS/FAIL checks only the finest aggregate electric and magnetic rates against a strict enhanced-order target. It does not certify all columns or all mesh transitions, and the electric target is not guaranteed on a generic unstructured tetrahedral family.

21 Summary

The convergence driver evaluates the degree- N DG solution at Jaskowiec–Sukumar volume cubature points, maps those points to each physical tetrahedron, compares all six numerical components with the analytical plane wave, and integrates squared errors using $|\det J_K| w_q$. MPI sums owned-element contributions globally.

Component norms are combined by Euclidean sums to produce electric, magnetic, and total field norms. Consecutive-level rates use the measured characteristic-size ratio and are written for aggregate and component errors. The automated verdict uses only the finest aggregate electric and magnetic rates, requiring at least $N + 1 - 0.5$ and $N - 0.5$, respectively. The console also reports the active E_z and H_y errors and transverse leakage, because the strict aggregate electric target can be reduced by order- N polarization leakage on unstructured tetrahedra.

The methodology is suitable for distributed h -convergence assessment provided that users also check temporal sensitivity, quadrature adequacy, pre-asymptotic mesh behavior, and the special interpretation required for exact-zero field components.